Where the Wind Carries Pollution: Evidence from Home-Based EMS Calls in New York City

Ives Xue

Advisor: Dr. Alex (Chu) Yu

Introduction

Where the Wind Carries Pollution

- Does **short-run wind-driven pollution exposure** affect **Emergency Medical Service (EMS) calls?**
- Focus: **NYC neighborhoods, 2013–2018**
- Compare **dirty-fuel power plants** vs. **interstate highways**

Why Study This?

- Air pollution burden is **uneven** across neighborhoods
- A **source-based approach** (e.g., Herrnstadt 2021, AEJ: Applied) better links emissions to **exposure** and **policy-relevant variation**
- Ambient monitor data are **too sparse** to capture neighborhood-level variation

Data Construction

Downwind Power Plants Construction

- Identify operating plants using EPA eGRID & Classify them as dirty or clean
- Match each plant to nearby NYC ZCTAs, then use daily wind direction to classify whether a ZCTA is downwind
- Create a daily indicator for plant exposure within **0.3 km, 1 km, 2 km**
- **Downwind Interstate Highways data is provided by Dr. Yu**

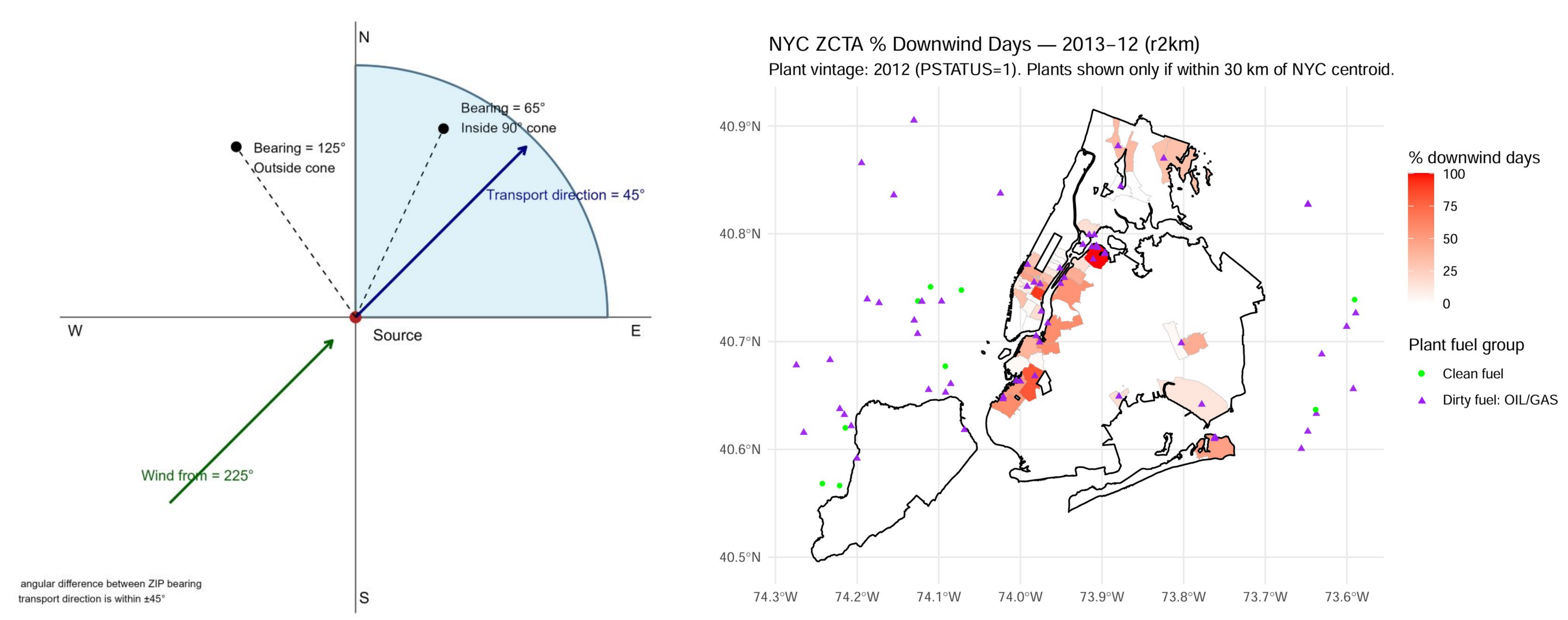

Figure 1&2: Downwind Power Plant Classification Rule and Percentage in December 2013

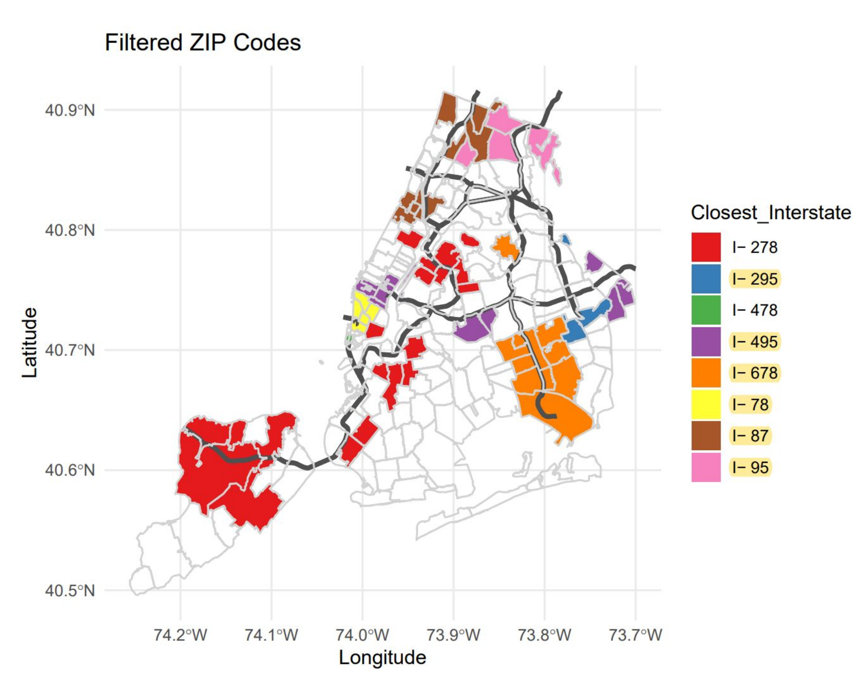

Figure 3: Zip Codes Within 1km of Interstates

Main Analysis Sample

Data Description

- **Unit of observation:** ZCTA-day panel for New York City, 2013–2018
- **Data Source:** Emergency Medical Service (**EMS**) Data from NYC Open Data, **weather data** from National Centers for Environmental Information (**NCEI**), and self-constructed **Downwind Data**.
- **Home dispatches**: Calls occurring between **7 p.m. and 7 a.m.**
- **Sick cases:** Dispatches not immediately assigned to a more specific case type
- **Rate construction:** For each ZCTA-day, EMS call rate = 100,000 × (number of dispatches / population)
- **Key outcomes: Home Sick-Case Rate** and **Home Total EMS Rate**
- **Key exposures:** Downwind Dirty-Fuel Plant (2km) and Downwind Interstate(90°)

Variable	N	Mean	SD	Min	Max
Panel A. Power-Plant Sample					
Outcomes					
Home Sick-Case Rate	82,817	3.206	12.020	0.000	2732.240
Home Total EMS Rate	82,817	19.904	54.690	0.906	7103.825
Main exposure					
Downwind Dirty-Fuel Plant (2 km)	82,817	0.344	0.475	0.000	1.000
Panel B. Interstate Sample					
Outcomes					
Home Sick-Case Rate	105,285	3.206	12.020	0.000	2732.240
Home Total EMS Rate	105,285	19.904	54.690	0.906	7103.825
Main exposure					
Downwind Interstate (90°)	105,285	0.243	0.429	0.000	1.000

Notes: Panel A reports descriptive statistics for the estimation sample used in the preferred power-plant specification. Panel B reports descriptive statistics for the estimation sample used in the preferred interstate specification. EMS outcomes are daily home-based call rates per 100,000 residents. "Home" refers to dispatches occurring between 7 p.m. and 7 a.m., when the recorded ZCTA is more likely to reflect a residential location. The exposure variables are binary indicators, so their means can be interpreted as the share of ZCTA-days classified as downwind under the corresponding preferred definition. Weather controls are used in the regressions but omitted here for brevity.

Table 1: Descriptive Statistics for the Main Estimation Samples

Neighborhood Differences

Proximity to Power Plants

- Neighborhoods **near** power plants **differ** systematically from those **farther away**
- This makes **raw** cross-sectional comparisons **difficult to interpret**
- **Regional and Time Fixed Effects** are needed

Variable	Within 0.5 km	0.5–2 km	Outside 2 km	Near-Far Gap
Below poverty (%)	28.361	16.337	14.288	14.074
Unemployed (%)	12.967	7.202	8.472	4.495
Minority (%)	70.191	39.988	58.590	11.601
Overcrowded renter (%)	4.408	3.400	3.192	1.216
Female single parent (%)	7.696	4.545	5.841	1.854
Black (%)	25.900	5.900	14.800	11.100
SNAP / Food stamps (%)	7.000	5.600	4.700	2.300
Cash public assistance (%)	1.700	1.200	1.100	0.600

Notes: This table reports median demographic characteristics for ZCTAs grouped into three mutually exclusive distance categories relative to nearby power plants: within 0.5 km, 0.5–2 km, and outside 2 km. All entries are percentages. The final column reports the difference between the median for ZCTAs within 0.5 km and the median for ZCTAs outside 2 km, so larger values indicate a clearer near-far contrast.

Table 2: Neighborhood Characteristics by Distance to Nearby Power Plants

Method & Result

Regressions

$$\textbf{Power-Plant Model:}\quad Y_{zt} = \beta_1\,\text{Downwind Dirty-Fuel Plant}_{zt} + \mathbf{W}'_{zt}\gamma + \alpha_z + \lambda_{ym(t)} + \delta_{dow(t)} + \varepsilon_{zt}$$

$$\textbf{Interstate Model:}\quad Y_{zt} = \beta_2\,\text{Downwind Interstate}_{zt} + \mathbf{W}'_{zt}\gamma + \alpha_z + \lambda_{ym(t)} + \delta_{dow(t)} + \varepsilon_{zt}$$

$$\begin{aligned}\textbf{Joint Model:}\quad Y_{zt} &= \beta_1\,\text{Downwind Dirty-Fuel Plant}_{zt} + \beta_2\,\text{Downwind Interstate}_{zt}\\ &+ \beta_3(\text{Downwind Dirty-Fuel Plant}_{zt} \times \text{Downwind Interstate}_{zt})\\ &+ \mathbf{W}'_{zt}\gamma + \alpha_z + \lambda_{ym(t)} + \delta_{dow(t)} + \varepsilon_{zt}\end{aligned}$$

- We estimate whether home-based EMS call rates increase when a ZCTA is downwind of a pollution source
- All models include daily weather controls, ZCTA fixed effects, year-month fixed effects, and day-of-week fixed effects

Main Result

	(1) Home Sick-Case Rate	(2) Home Total EMS Rate
Panel A. Main Power-Plant Specification		
Downwind Dirty-Fuel Plant (2 km)	0.0448**	0.0211
	(0.0201)	(0.0626)
Observations	82,817	81,718
R^2	0.3212	0.7231
Panel B. Main Interstate Specification		
Downwind Interstate (90°)	-0.0121	-0.0670
	(0.0205)	(0.0507)
Observations	105,285	101,855
R^2	0.2775	0.6666
Panel C. Joint Specification (60° Interstate Measure)		
Downwind Dirty-Fuel Plant (2 km)	0.105**	-0.060
	(0.045)	(0.163)
Downwind Interstate (60°)	0.149**	0.091
	(0.066)	(0.189)
Downwind Dirty-Fuel Plant × Downwind Interstate	-0.279**	-0.325
	(0.112)	(0.337)
Observations	33,301	32,802
R^2	0.3020	0.7200
Weather controls	Yes	Yes
Population weights	Yes	Yes
Clustered SE	ZCTA	ZCTA

Notes: All dependent variables are home-based daily EMS call rates per 100,000 residents at the ZCTA level. Home-based refers to dispatches occurring between 7 p.m. and 7 a.m., when the recorded ZCTA is more likely to reflect a residential location. Column (1) reports the Home Sick-Case Rate, and column (2) reports the Home Total EMS Rate. Panel A reports the preferred power-plant specification using the downwind dirty-fuel plant indicator within 2 km. Panel B reports the preferred interstate specification using the downwind interstate indicator defined by a 90° cone. Panel C reports a joint specification that uses the 60° interstate measure, the downwind dirty-fuel plant indicator within 2 km, and their interaction. All regressions include weather controls, ZCTA fixed effects, year-month fixed effects, day-of-week fixed effects, and population weights. Standard errors clustered at the ZCTA level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Baseline Fixed-Effects Results for Home-Based EMS Call Rates

Other Models

Besides three models above, we also have checked the following models:

- Models use PUMA as regional fixed effects
- Models use Respiratory Case Rate as main outcome
- Models with socioeconomic controls